

with one that incorporates prior knowledge, namely,

$$\sum_i \ln \left(1 + e^{-y_i f(x_i)} \right) + \eta \sum_i \text{RE} \left(p(x_i) \parallel \frac{1}{1 + e^{-f(x_i)}} \right)$$

where $\text{RE}(p \parallel q) = p \ln(p/q) + (1 - p) \ln((1 - p)/(1 - q))$ is binary relative entropy. The first term is the same as that in Eq. (10). The second term gives a measure of the distance from the model built by boosting to the human’s model. Thus, we balance the conditional likelihood of the data against the distance from our model to the human’s model. The relative importance of the two terms is controlled by the parameter η .

9 Experiments and applications

Practically, AdaBoost has many advantages. It is fast, simple and easy to program. It has no parameters to tune (except for the number of round T). It requires no prior knowledge about the base learner and so can be flexibly combined with *any* method for finding base classifiers. Finally, it comes with a set of theoretical guarantees given sufficient data and a base learner that can reliably provide only moderately accurate base classifiers. This is a shift in mind set for the learning-system designer: instead of trying to design a learning algorithm that is accurate over the entire space, we can instead focus on finding base learning algorithms that only need to be better than random.

On the other hand, some caveats are certainly in order. The actual performance of boosting on a particular problem is clearly dependent on the data and the base learner. Consistent with theory, boosting can fail to perform well given insufficient data, overly complex base classifiers or base classifiers that are too weak. Boosting seems to be especially susceptible to noise [18] (more on this in Sectionsec:exps).

AdaBoost has been tested empirically by many researchers, including [4, 18, 21, 40, 49, 59, 73]. For instance, Freund and Schapire [30] tested AdaBoost on a set of UCI benchmark datasets [54] using C4.5 [60] as a base learning algorithm, as well as an algorithm that finds the best “decision stump” or single-test decision tree. Some of the results of these experiments are shown in Fig. 3. As can be seen from this figure, even boosting the weak decision stumps can usually give as good results as C4.5, while boosting C4.5 generally gives the decision-tree algorithm a significant improvement in performance.

In another set of experiments, Schapire and Singer [71] used boosting for text categorization tasks. For this work, base classifiers were used that test on the presence or absence of a word or phrase. Some results of these experiments comparing

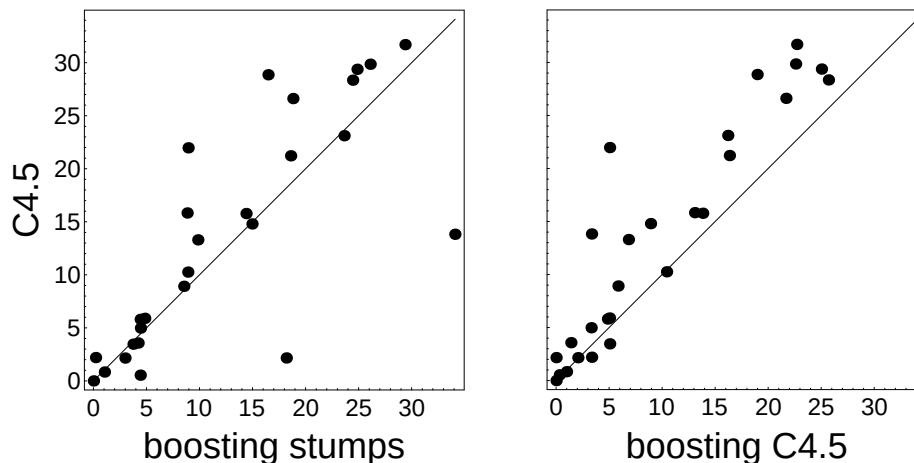


Figure 3: Comparison of C4.5 versus boosting stumps and boosting C4.5 on a set of 27 benchmark problems as reported by Freund and Schapire [30]. Each point in each scatterplot shows the test error rate of the two competing algorithms on a single benchmark. The y -coordinate of each point gives the test error rate (in percent) of C4.5 on the given benchmark, and the x -coordinate gives the error rate of boosting stumps (left plot) or boosting C4.5 (right plot). All error rates have been averaged over multiple runs.

AdaBoost to four other methods are shown in Fig. 4. In nearly all of these experiments and for all of the performance measures tested, boosting performed as well or significantly better than the other methods tested. As shown in Fig. 5, these experiments also demonstrated the effectiveness of using confidence-rated predictions [70], mentioned in Section 3 as a means of speeding up boosting.

Boosting has also been applied to text filtering [72] and routing [39], “ranking” problems [28], learning problems arising in natural language processing [1, 12, 25, 38, 55, 78], image retrieval [74], medical diagnosis [53], and customer monitoring and segmentation [56, 57].

Rochery et al.’s [64, 65] method of incorporating human knowledge into boosting, described in Section 8, was applied to two speech categorization tasks. In this case, the prior knowledge took the form of a set of hand-built rules mapping keywords to predicted categories. The results are shown in Fig. 6.

The final classifier produced by AdaBoost when used, for instance, with a decision-tree base learning algorithm, can be extremely complex and difficult to comprehend. With greater care, a more human-understandable final classifier can be obtained using boosting. Cohen and Singer [11] showed how to design a base

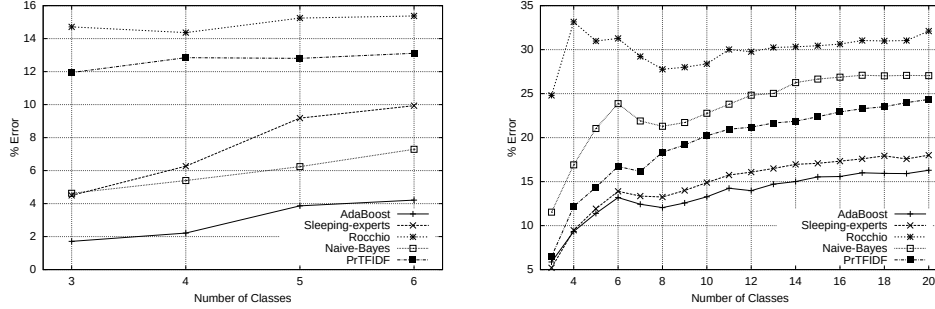


Figure 4: Comparison of error rates for AdaBoost and four other text categorization methods (naive Bayes, probabilistic TF-IDF, Rocchio and sleeping experts) as reported by Schapire and Singer [71]. The algorithms were tested on two text corpora — Reuters newswire articles (left) and AP newswire headlines (right) — and with varying numbers of class labels as indicated on the x -axis of each figure.

learning algorithm that, when combined with AdaBoost, results in a final classifier consisting of a relatively small set of rules similar to those generated by systems like RIPPER [10], IREP [36] and C4.5rules [60]. Cohen and Singer’s system, called SLIPPER, is fast, accurate and produces quite compact rule sets. In other work, Freund and Mason [29] showed how to apply boosting to learn a generalization of decision trees called “alternating trees.” Their algorithm produces a single alternating tree rather than an ensemble of trees as would be obtained by running AdaBoost on top of a decision-tree learning algorithm. On the other hand, their learning algorithm achieves error rates comparable to those of a whole ensemble of trees.

A nice property of AdaBoost is its ability to identify *outliers*, i.e., examples that are either mislabeled in the training data, or that are inherently ambiguous and hard to categorize. Because AdaBoost focuses its weight on the hardest examples, the examples with the highest weight often turn out to be outliers. An example of this phenomenon can be seen in Fig. 7 taken from an OCR experiment conducted by Freund and Schapire [30].

When the number of outliers is very large, the emphasis placed on the hard examples can become detrimental to the performance of AdaBoost. This was demonstrated very convincingly by Dietterich [18]. Friedman, Hastie and Tibshirani [34] suggested a variant of AdaBoost, called “Gentle AdaBoost” that puts less emphasis on outliers. Rätsch, Onoda and Müller [61] show how to regularize AdaBoost to handle noisy data. Freund [27] suggested another algorithm, called “BrownBoost,” that takes a more radical approach that de-emphasizes outliers when it seems clear that they are “too hard” to classify correctly. This algorithm, which is an adaptive

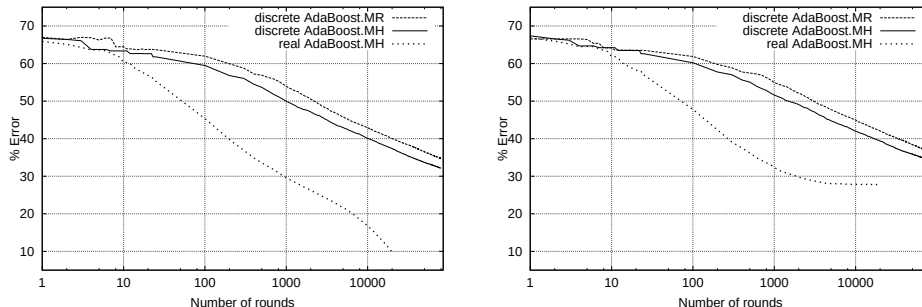


Figure 5: Comparison of the training (left) and test (right) error using three boosting methods on a six-class text classification problem from the TREC-AP collection, as reported by Schapire and Singer [70, 71]. Discrete AdaBoost.MH and discrete AdaBoost.MR are multiclass versions of AdaBoost that require binary ($\{-1, +1\}$ -valued) base classifiers, while real AdaBoost.MH is a multiclass version that uses “confidence-rated” (i.e., real-valued) base classifiers.

version of Freund’s [26] “boost-by-majority” algorithm, demonstrates an intriguing connection between boosting and Brownian motion.

10 Conclusion

In this overview, we have seen that there have emerged a great many views or interpretations of AdaBoost. First and foremost, AdaBoost is a genuine boosting algorithm: given access to a true weak learning algorithm that always performs a little bit better than random guessing on every distribution over the training set, we can prove arbitrarily good bounds on the training error and generalization error of AdaBoost.

Besides this original view, AdaBoost has been interpreted as a procedure based on functional gradient descent, as an approximation of logistic regression and as a repeated-game playing algorithm. AdaBoost has also been shown to be related to many other topics, such as game theory and linear programming, Bregman distances, support-vector machines, Brownian motion, logistic regression and maximum-entropy methods such as iterative scaling.

All of these connections and interpretations have greatly enhanced our understanding of boosting and contributed to its extension in ever more practical directions, such as to logistic regression and other loss-minimization problems, to multiclass problems, to incorporate regularization and to allow the integration of prior background knowledge.

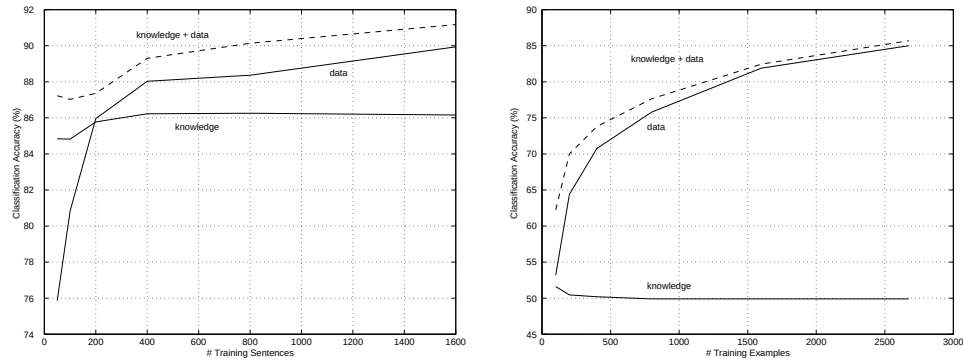


Figure 6: Comparison of percent classification accuracy on two spoken language tasks (“How may I help you” on the left and “Help desk” on the right) as a function of the number of training examples using data and knowledge separately or together, as reported by Rochery et al. [64, 65].

We also have discussed a few of the growing number of applications of AdaBoost to practical machine learning problems, such as text and speech categorization.

References

- [1] Steven Abney, Robert E. Schapire, and Yoram Singer. Boosting applied to tagging and PP attachment. In *Proceedings of the Joint SIGDAT Conference on Empirical Methods in Natural Language Processing and Very Large Corpora*, 1999.
- [2] Erin L. Allwein, Robert E. Schapire, and Yoram Singer. Reducing multiclass to binary: A unifying approach for margin classifiers. *Journal of Machine Learning Research*, 1:113–141, 2000.
- [3] Peter L. Bartlett. The sample complexity of pattern classification with neural networks: the size of the weights is more important than the size of the network. *IEEE Transactions on Information Theory*, 44(2):525–536, March 1998.
- [4] Eric Bauer and Ron Kohavi. An empirical comparison of voting classification algorithms: Bagging, boosting, and variants. *Machine Learning*, 36(1/2):105–139, 1999.
- [5] Eric B. Baum and David Haussler. What size net gives valid generalization? *Neural Computation*, 1(1):151–160, 1989.
- [6] Anselm Blumer, Andrzej Ehrenfeucht, David Haussler, and Manfred K. Warmuth. Learnability and the Vapnik-Chervonenkis dimension. *Journal of the Association for Computing Machinery*, 36(4):929–965, October 1989.

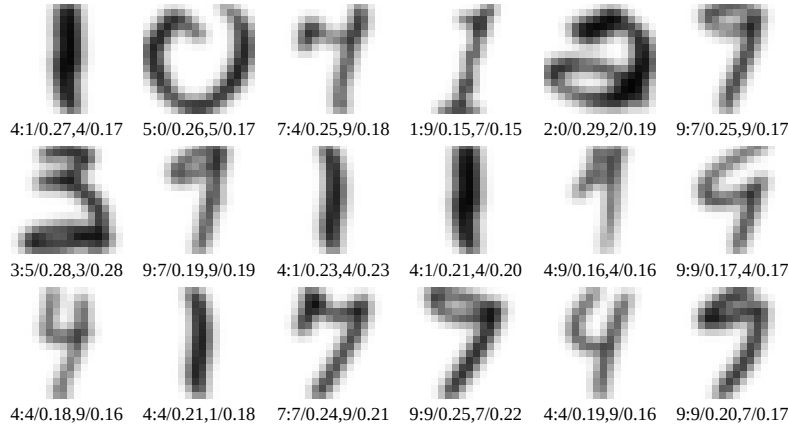


Figure 7: A sample of the examples that have the largest weight on an OCR task as reported by Freund and Schapire [30]. These examples were chosen after 4 rounds of boosting (top line), 12 rounds (middle) and 25 rounds (bottom). Underneath each image is a line of the form $d:\ell_1/w_1,\ell_2/w_2$, where d is the label of the example, ℓ_1 and ℓ_2 are the labels that get the highest and second highest vote from the combined classifier at that point in the run of the algorithm, and w_1, w_2 are the corresponding normalized scores.

- [7] Bernhard E. Boser, Isabelle M. Guyon, and Vladimir N. Vapnik. A training algorithm for optimal margin classifiers. In *Proceedings of the Fifth Annual ACM Workshop on Computational Learning Theory*, pages 144–152, 1992.
- [8] Leo Breiman. Arcing classifiers. *The Annals of Statistics*, 26(3):801–849, 1998.
- [9] Leo Breiman. Prediction games and arcing classifiers. *Neural Computation*, 11(7):1493–1517, 1999.
- [10] William Cohen. Fast effective rule induction. In *Proceedings of the Twelfth International Conference on Machine Learning*, pages 115–123, 1995.
- [11] William W. Cohen and Yoram Singer. A simple, fast, and effective rule learner. In *Proceedings of the Sixteenth National Conference on Artificial Intelligence*, 1999.
- [12] Michael Collins. Discriminative reranking for natural language parsing. In *Proceedings of the Seventeenth International Conference on Machine Learning*, 2000.
- [13] Michael Collins, Robert E. Schapire, and Yoram Singer. Logistic regression, Ada-Boost and Bregman distances. *Machine Learning*, to appear.
- [14] Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine Learning*, 20(3):273–297, September 1995.

- [15] J. N. Darroch and D. Ratcliff. Generalized iterative scaling for log-linear models. *The Annals of Mathematical Statistics*, 43(5):1470–1480, 1972.
- [16] Stephen Della Pietra, Vincent Della Pietra, and John Lafferty. Inducing features of random fields. *IEEE Transactions Pattern Analysis and Machine Intelligence*, 19(4):1–13, April 1997.
- [17] Ayhan Demiriz, Kristin P. Bennett, and John Shawe-Taylor. Linear programming boosting via column generation. *Machine Learning*, 46(1/2/3):225–254, 2002.
- [18] Thomas G. Dietterich. An experimental comparison of three methods for constructing ensembles of decision trees: Bagging, boosting, and randomization. *Machine Learning*, 40(2):139–158, 2000.
- [19] Thomas G. Dietterich and Ghulum Bakiri. Solving multiclass learning problems via error-correcting output codes. *Journal of Artificial Intelligence Research*, 2:263–286, January 1995.
- [20] Harris Drucker. Improving regressors using boosting techniques. In *Machine Learning: Proceedings of the Fourteenth International Conference*, pages 107–115, 1997.
- [21] Harris Drucker and Corinna Cortes. Boosting decision trees. In *Advances in Neural Information Processing Systems 8*, pages 479–485, 1996.
- [22] Harris Drucker, Robert Schapire, and Patrice Simard. Boosting performance in neural networks. *International Journal of Pattern Recognition and Artificial Intelligence*, 7(4):705–719, 1993.
- [23] Nigel Duffy and David Helmbold. Potential boosters? In *Advances in Neural Information Processing Systems 11*, 1999.
- [24] Nigel Duffy and David Helmbold. Boosting methods for regression. *Machine Learning*, 49(2/3), 2002.
- [25] Gerard Escudero, Lluís Màrquez, and German Rigau. Boosting applied to word sense disambiguation. In *Proceedings of the 12th European Conference on Machine Learning*, pages 129–141, 2000.
- [26] Yoav Freund. Boosting a weak learning algorithm by majority. *Information and Computation*, 121(2):256–285, 1995.
- [27] Yoav Freund. An adaptive version of the boost by majority algorithm. *Machine Learning*, 43(3):293–318, June 2001.
- [28] Yoav Freund, Raj Iyer, Robert E. Schapire, and Yoram Singer. An efficient boosting algorithm for combining preferences. In *Machine Learning: Proceedings of the Fifteenth International Conference*, 1998.
- [29] Yoav Freund and Llew Mason. The alternating decision tree learning algorithm. In *Machine Learning: Proceedings of the Sixteenth International Conference*, pages 124–133, 1999.

- [30] Yoav Freund and Robert E. Schapire. Experiments with a new boosting algorithm. In *Machine Learning: Proceedings of the Thirteenth International Conference*, pages 148–156, 1996.
- [31] Yoav Freund and Robert E. Schapire. Game theory, on-line prediction and boosting. In *Proceedings of the Ninth Annual Conference on Computational Learning Theory*, pages 325–332, 1996.
- [32] Yoav Freund and Robert E. Schapire. A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of Computer and System Sciences*, 55(1):119–139, August 1997.
- [33] Yoav Freund and Robert E. Schapire. Adaptive game playing using multiplicative weights. *Games and Economic Behavior*, 29:79–103, 1999.
- [34] Jerome Friedman, Trevor Hastie, and Robert Tibshirani. Additive logistic regression: A statistical view of boosting. *The Annals of Statistics*, 38(2):337–374, April 2000.
- [35] Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5), October 2001.
- [36] Johannes Fürnkranz and Gerhard Widmer. Incremental reduced error pruning. In *Machine Learning: Proceedings of the Eleventh International Conference*, pages 70–77, 1994.
- [37] Adam J. Grove and Dale Schuurmans. Boosting in the limit: Maximizing the margin of learned ensembles. In *Proceedings of the Fifteenth National Conference on Artificial Intelligence*, 1998.
- [38] Masahiko Haruno, Satoshi Shirai, and Yoshifumi Ooyama. Using decision trees to construct a practical parser. *Machine Learning*, 34:131–149, 1999.
- [39] Raj D. Iyer, David D. Lewis, Robert E. Schapire, Yoram Singer, and Amit Singhal. Boosting for document routing. In *Proceedings of the Ninth International Conference on Information and Knowledge Management*, 2000.
- [40] Jeffrey C. Jackson and Mark W. Craven. Learning sparse perceptrons. In *Advances in Neural Information Processing Systems 8*, pages 654–660, 1996.
- [41] Michael Kearns and Leslie G. Valiant. Learning Boolean formulae or finite automata is as hard as factoring. Technical Report TR-14-88, Harvard University Aiken Computation Laboratory, August 1988.
- [42] Michael Kearns and Leslie G. Valiant. Cryptographic limitations on learning Boolean formulae and finite automata. *Journal of the Association for Computing Machinery*, 41(1):67–95, January 1994.
- [43] Jyrki Kivinen and Manfred K. Warmuth. Boosting as entropy projection. In *Proceedings of the Twelfth Annual Conference on Computational Learning Theory*, pages 134–144, 1999.

- [44] V. Koltchinskii and D. Panchenko. Empirical margin distributions and bounding the generalization error of combined classifiers. *The Annals of Statistics*, 30(1), February 2002.
- [45] Vladimir Koltchinskii, Dmitriy Panchenko, and Fernando Lozano. Further explanation of the effectiveness of voting methods: The game between margins and weights. In *Proceedings 14th Annual Conference on Computational Learning Theory and 5th European Conference on Computational Learning Theory*, pages 241–255, 2001.
- [46] Vladimir Koltchinskii, Dmitriy Panchenko, and Fernando Lozano. Some new bounds on the generalization error of combined classifiers. In *Advances in Neural Information Processing Systems 13*, 2001.
- [47] John Lafferty. Additive models, boosting and inference for generalized divergences. In *Proceedings of the Twelfth Annual Conference on Computational Learning Theory*, pages 125–133, 1999.
- [48] Guy Lebanon and John Lafferty. Boosting and maximum likelihood for exponential models. In *Advances in Neural Information Processing Systems 14*, 2002.
- [49] Richard Maclin and David Opitz. An empirical evaluation of bagging and boosting. In *Proceedings of the Fourteenth National Conference on Artificial Intelligence*, pages 546–551, 1997.
- [50] Llew Mason, Peter Bartlett, and Jonathan Baxter. Direct optimization of margins improves generalization in combined classifiers. In *Advances in Neural Information Processing Systems 12*, 2000.
- [51] Llew Mason, Jonathan Baxter, Peter Bartlett, and Marcus Frean. Functional gradient techniques for combining hypotheses. In Alexander J. Smola, Peter J. Bartlett, Bernhard Schölkopf, and Dale Schuurmans, editors, *Advances in Large Margin Classifiers*. MIT Press, 1999.
- [52] Llew Mason, Jonathan Baxter, Peter Bartlett, and Marcus Frean. Boosting algorithms as gradient descent. In *Advances in Neural Information Processing Systems 12*, 2000.
- [53] Stefano Merler, Cesare Furlanello, Barbara Larcher, and Andrea Sboner. Tuning cost-sensitive boosting and its application to melanoma diagnosis. In *Multiple Classifier Systems: Proceedings of the 2nd International Workshop*, pages 32–42, 2001.
- [54] C. J. Merz and P. M. Murphy. UCI repository of machine learning databases, 1999. www.ics.uci.edu/~mllearn/MLRepository.html.
- [55] Pedro J. Moreno, Beth Logan, and Bhiksha Raj. A boosting approach for confidence scoring. In *Proceedings of the 7th European Conference on Speech Communication and Technology*, 2001.
- [56] Michael C. Mozer, Richard Wolniewicz, David B. Grimes, Eric Johnson, and Howard Kaushansky. Predicting subscriber dissatisfaction and improving retention in the wireless telecommunications industry. *IEEE Transactions on Neural Networks*, 11:690–696, 2000.

- [57] Takashi Onoda, Gunnar Rätsch, and Klaus-Robert Müller. Applying support vector machines and boosting to a non-intrusive monitoring system for household electric appliances with inverters. In *Proceedings of the Second ICSC Symposium on Neural Computation*, 2000.
- [58] Dmitriy Panchenko. New zero-error bounds for voting algorithms. Unpublished manuscript, 2001.
- [59] J. R. Quinlan. Bagging, boosting, and C4.5. In *Proceedings of the Thirteenth National Conference on Artificial Intelligence*, pages 725–730, 1996.
- [60] J. Ross Quinlan. *C4.5: Programs for Machine Learning*. Morgan Kaufmann, 1993.
- [61] G. Rätsch, T. Onoda, and K.-R. Müller. Soft margins for AdaBoost. *Machine Learning*, 42(3):287–320, 2001.
- [62] Gunnar Rätsch, Manfred Warmuth, Sebastian Mika, Takashi Onoda, Steven Lemm, and Klaus-Robert Müller. Barrier boosting. In *Proceedings of the Thirteenth Annual Conference on Computational Learning Theory*, pages 170–179, 2000.
- [63] Greg Ridgeway, David Madigan, and Thomas Richardson. Boosting methodology for regression problems. In *Proceedings of the International Workshop on AI and Statistics*, pages 152–161, 1999.
- [64] M. Rochery, R. Schapire, M. Rahim, N. Gupta, G. Riccardi, S. Bangalore, H. Alshawi, and S. Douglas. Combining prior knowledge and boosting for call classification in spoken language dialogue. Unpublished manuscript, 2001.
- [65] Marie Rochery, Robert Schapire, Mazin Rahim, and Narendra Gupta. BoosTexter for text categorization in spoken language dialogue. Unpublished manuscript, 2001.
- [66] Robert E. Schapire. The strength of weak learnability. *Machine Learning*, 5(2):197–227, 1990.
- [67] Robert E. Schapire. Using output codes to boost multiclass learning problems. In *Machine Learning: Proceedings of the Fourteenth International Conference*, pages 313–321, 1997.
- [68] Robert E. Schapire. Drifting games. *Machine Learning*, 43(3):265–291, June 2001.
- [69] Robert E. Schapire, Yoav Freund, Peter Bartlett, and Wee Sun Lee. Boosting the margin: A new explanation for the effectiveness of voting methods. *The Annals of Statistics*, 26(5):1651–1686, October 1998.
- [70] Robert E. Schapire and Yoram Singer. Improved boosting algorithms using confidence-rated predictions. *Machine Learning*, 37(3):297–336, December 1999.
- [71] Robert E. Schapire and Yoram Singer. BoosTexter: A boosting-based system for text categorization. *Machine Learning*, 39(2/3):135–168, May/June 2000.
- [72] Robert E. Schapire, Yoram Singer, and Amit Singhal. Boosting and Rocchio applied to text filtering. In *Proceedings of the 21st Annual International Conference on Research and Development in Information Retrieval*, 1998.

- [73] Holger Schwenk and Yoshua Bengio. Training methods for adaptive boosting of neural networks. In *Advances in Neural Information Processing Systems 10*, pages 647–653, 1998.
- [74] Kinh Tieu and Paul Viola. Boosting image retrieval. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2000.
- [75] L. G. Valiant. A theory of the learnable. *Communications of the ACM*, 27(11):1134–1142, November 1984.
- [76] V. N. Vapnik and A. Ya. Chervonenkis. On the uniform convergence of relative frequencies of events to their probabilities. *Theory of Probability and its applications*, XVI(2):264–280, 1971.
- [77] Vladimir N. Vapnik. *The Nature of Statistical Learning Theory*. Springer, 1995.
- [78] Marilyn A. Walker, Owen Rambow, and Monica Rogati. SPoT: A trainable sentence planner. In *Proceedings of the 2nd Annual Meeting of the North American Chapter of the Association for Computational Linguistics*, 2001.